

CHAPTER C7

SNOW LOADS

Methodology. The procedure established for determining design snow loads is as follows:

1. Determine the ground snow load for the geographic location (Sections 7.2 and C7.2).
2. Generate a flat roof snow load from the ground load with consideration given to (1) roof exposure (Sections 7.3.1, C7.3, and C7.3.1); (2) roof thermal condition (Sections 7.3.2, C7.3, and C7.3.2); and (3) occupancy and function of structure (Sections 7.3.3 and C7.3.3).
3. Consider roof slope (Sections 7.4 through 7.4.5 and C7.4).
4. Consider partial loading (Sections 7.5 and C7.5).
5. Consider unbalanced loads (Sections 7.6 through 7.6.4 and C7.6).
6. Consider snow drifts: (1) on lower roofs (Sections 7.7 through 7.7.2 and C7.7) and (2) from projections (Sections 7.8 and C7.8).
7. Consider sliding snow (Sections 7.9 and C7.9).
8. Consider extra loads from rain on snow (Sections 7.10 and C7.10).
9. Consider ponding loads (Section 7.11 and C7.11).
10. Consider existing roofs (Sections 7.12 and C7.12).
11. Consider other roofs and sites (Section C7.13).
12. Consider the consequences of loads in excess of the design value (see the following text).

Loads in Excess of the Design Value. The philosophy of the probabilistic approach used in this standard is to establish a design value that reduces the risk of a snow load-induced failure to an acceptably low level. Because snow loads in excess of the design value may occur, the implications of such “excess” loads should be considered. For example, if a roof is deflected at the design snow load so that slope to drain is eliminated, “excess” snow load might cause ponding (Section C7.11) and perhaps progressive failure.

The snow load/dead load ratio of a roof structure is an important consideration when assessing the implications of “excess” loads. If the design snow load is exceeded, the percentage increase in total load would be greater for a lightweight structure (i.e., one with a high snow load/dead load ratio) than for a heavy structure (i.e., one with a low snow load/dead load ratio). For example, if a 40 lb/ft² (1.92 kN/m²) roof snow load is exceeded by 20 lb/ft² (0.96 kN/m²) for a roof that has a 25 lb/ft² (1.19 kN/m²) dead load, the total load increases by 31% from 65 to 85 lb/ft² (3.11 to 4.07 kN/m²). If the roof had a 60 lb/ft² (2.87 kN/m²) dead load, the total load would increase only by 20% from 100 to 120 lb/ft² (4.79 to 5.75 kN/m²).

C7.2 GROUND SNOW LOADS, p_g

The mapped snow load provisions in Fig. 7.2-1 were developed from an extreme-value statistical analysis of weather records of snow on the ground (Ellingwood and Redfield 1983). The map was produced by the Corps of Engineers, Cold Regions Research and Engineering Laboratory (CRREL). The lognormal distribution was selected to estimate ground snow loads, which have a 2% annual probability of being exceeded (50-yr mean recurrence interval).

Maximum measured ground snow loads and ground snow loads with a 2% annual probability of being exceeded are presented in Table C7.2-1 for National Weather Service (NWS) “first-order” stations at which ground snow loads have been measured for at least 11 years during the period 1952–1992.

Concurrent records of the depth and load of snow on the ground at 204 NWS first-order stations were used to estimate the ground snow load and the ground snow depth that has a 2% annual probability of being exceeded for each of these locations. The period of record for these 204 locations, where both snow depth and snow load have been measured, averages 33 years up through the winter of 1991–1992. A mathematical relationship was developed between the 2% depths and the 2% loads. The nonlinear best-fit relationship between these extreme values was used to estimate 2% (50-yr mean recurrence interval) ground snow loads at about 9,200 other locations at which only snow depths were measured. These loads, as well as the extreme-value loads developed directly from snow load measurements at 204 first-order locations, were used to construct the maps.

In general, loads from these two sources were in agreement. In areas where there were differences, loads from the 204 first-order locations were considered to be more valuable when the map was constructed. This procedure ensures that the map is referenced to the NWS observed loads and contains spatial detail provided by snow depth measurements at about 9,200 other locations.

The maps were generated from data current through the 1991–1992 winter. Where statistical studies using more recent information are available, they may be used to produce improved design guidance.

However, adding a big snow year to data developed from periods of record exceeding 20 years will usually not change 50-yr values much. As examples, the databases for Boston and Chattanooga, Tennessee, were updated to include the winters of 1992–1993 and 1993–1994 because record snows occurred there during that period. In Boston, 50-yr loads based on water equivalent measurements only increased from 34 to 35 lb/ft² (1.63 to 1.68 kN/m²), and loads generated from snow depth measurements remained at 25 lb/ft² (1.20 kN/m²). In Chattanooga, loads generated from water equivalent measurements increased from 6 to 7 lb/ft² (0.29 to 0.34 kN/m²), and loads

Table C7.2-1 Ground Snow Loads at National Weather Service Locations Where Load Measurements Are Made

Location	Ground Snow Load (lb/ft ²)		
	Years of Record	Maximum Observed	2% Annual Probability ^a
ALABAMA			
Birmingham	40	4	3
Huntsville	33	7	5
Mobile	40	1	1
ARIZONA			
Flagstaff	38	88	48
Tucson	40	3	3
Winslow	39	12	7
ARKANSAS			
Fort Smith	37	6	5
Little Rock	24	6	6
CALIFORNIA			
Bishop	31	6	8
Blue Canyon	26	213	242
Mt. Shasta	32	62	62
Red Bluff	34	3	3
COLORADO			
All cities	NA	NA	See Table 7.2-2
CONNECTICUT			
Bridgeport	39	21	24
Hartford	40	23	33
New Haven	17	11	15
DELAWARE			
Wilmington	39	12	16
GEORGIA			
Athens	40	6	5
Atlanta	39	4	3
Augusta	40	8	7
Columbus	39	1	1
Macon	40	8	7
Rome	28	3	3
IDAHO			
All cities	NA	NA	See Table 7.2-3
ILLINOIS			
Chicago	26	37	22
Chicago-O'Hare	32	25	17
Moline	39	21	19
Peoria	39	27	15
Rockford	26	31	19
Springfield	40	20	21
INDIANA			
Evansville	40	12	17
Fort Wayne	40	23	20
Indianapolis	40	19	22
South Bend	39	58	41
IOWA			
Burlington	11	15	17
Des Moines	40	22	22
Dubuque	39	34	32
Sioux City	38	28	28
Waterloo	33	25	32
KANSAS			
Concordia	30	12	17
Dodge City	40	10	14
Goodland	39	12	15
Topeka	40	18	17
Wichita	40	10	14
KENTUCKY			
Covington	40	22	13
Jackson	11	12	18

Table C7.2-1 (Continued)

Location	Ground Snow Load (lb/ft ²)		
	Years of Record	Maximum Observed	2% Annual Probability ^a
Lexington	40	15	13
Louisville	39	11	12
LOUISIANA			
Alexandria	17	2	2
Shreveport	40	4	3
MAINE			
Caribou	34	68	95
Portland	39	51	60
MARYLAND			
Baltimore	40	20	22
MASSACHUSETTS			
Boston	39	25	34
Nantucket	16	14	24
Worcester	33	29	44
MICHIGAN			
Alpena	31	34	48
Detroit Airport	34	27	18
Detroit City	14	6	10
Detroit—Willow	12	11	22
Flint	37	20	24
Grand Rapids	40	32	36
Houghton Lake	28	33	48
Lansing	35	34	36
Marquette	16	44	53
Muskegon	40	40	51
Sault Ste. Marie	40	68	77
MINNESOTA			
Duluth	40	55	63
International Falls	40	43	44
Minneapolis-St. Paul	40	34	51
Rochester	40	30	47
St. Cloud	40	40	53
MISSISSIPPI			
Jackson	40	3	3
Meridian	39	2	2
MISSOURI			
Columbia	39	19	20
Kansas City	40	18	18
St. Louis	37	28	21
Springfield	39	14	14
MONTANA			
All cities	NA	NA	See Table 7.2-4
NEBRASKA			
Grand Island	40	24	23
Lincoln	20	15	22
Norfolk	40	28	25
North Platte	39	16	13
Omaha	25	23	20
Scottsbluff	40	10	12
Valentine	26	26	22
NEVADA			
Elko	12	12	20
Ely	40	10	9
Las Vegas	39	3	3
Reno	39	12	11
Winnemucca	39	7	7
NEW HAMPSHIRE			
All cities	NA	NA	See Table 7.2-8
NEW JERSEY			
Atlantic City	35	12	15
Newark	39	18	15

continues

Table C7.2-1 (Continued)

Location	Ground Snow Load (lb/ft ²)		
	Years of Record	Maximum Observed	2% Annual Probability ^a
NEW MEXICO			
All cities	NA	NA	See Table 7.2-6
NEW YORK			
Albany	40	26	27
Binghamton	40	30	35
Buffalo	40	41	39
NYC–Kennedy	18	8	15
NYC–LaGuardia	40	23	16
Rochester	40	33	38
Syracuse	40	32	32
NORTH CAROLINA			
Asheville	28	7	14
Cape Hatteras	34	5	5
Charlotte	40	8	11
Greensboro	40	14	11
Raleigh-Durham	36	13	14
Wilmington	39	14	7
Winston-Salem	12	14	20
NORTH DAKOTA			
Bismarck	40	27	27
Fargo	39	27	41
Williston	40	28	27
OHIO			
Akron-Canton	40	16	14
Cleveland	40	27	19
Columbus	40	11	11
Dayton	40	18	11
Mansfield	30	31	17
Toledo Express	36	10	10
Youngstown	40	14	10
OKLAHOMA			
Oklahoma City	40	10	8
Tulsa	40	5	8
OREGON			
All cities	NA	NA	See Table 7.2-7
PENNSYLVANIA			
Allentown	40	16	23
Erie	32	20	18
Harrisburg	19	21	23
Philadelphia	39	13	14
Pittsburgh	40	27	20
Scranton	37	13	18
Williamsport	40	18	21
RHODE ISLAND			
Providence	39	22	23
SOUTH CAROLINA			
Charleston	39	2	2
Columbia	38	9	8
Florence	23	3	3
Greenville-Spartanburg	24	6	7
SOUTH DAKOTA			
Aberdeen	27	23	43
Huron	40	41	46
Rapid City	40	14	15
Sioux Falls	39	40	40
TENNESSEE			
Bristol	40	7	9
Chattanooga	40	6	6
Knoxville	40	10	9

Table C7.2-1 (Continued)

Location	Ground Snow Load (lb/ft ²)		
	Years of Record	Maximum Observed	2% Annual Probability ^a
Memphis	40	7	6
Nashville	40	6	9
TEXAS			
Abilene	40	6	6
Amarillo	39	15	10
Austin	39	2	2
Dallas	23	3	3
El Paso	38	8	8
Fort Worth	39	5	4
Lubbock	40	9	11
Midland	38	4	4
San Angelo	40	3	3
San Antonio	40	9	4
Waco	40	3	2
Wichita Falls	40	4	5
UTAH			
Milford	23	23	14
Salt Lake City	40	11	11
Wendover	13	2	3
VERMONT			
Burlington	40	43	36
VIRGINIA			
Dulles Airport	29	15	23
Lynchburg	40	13	18
National Airport	40	16	22
Norfolk	38	9	10
Richmond	40	11	16
Roanoke	40	14	20
WASHINGTON			
All cities	NA	NA	Table 7.2-5
WEST VIRGINIA			
Beckley	20	20	30
Charleston	38	21	18
Elkins	32	22	18
Huntington	30	15	19
WISCONSIN			
Green Bay	40	37	36
La Crosse	16	23	32
Madison	40	32	35
Milwaukee	40	34	29
WYOMING			
Casper	40	9	10
Cheyenne	40	18	18
Lander	39	26	24
Sheridan	40	20	23

^aIt is not appropriate to use only the site-specific information in this table for design purposes. Reasons are given in Section C7.2.

Note: To convert lb/ft² to kN/m², multiply by 0.0479.

generated from snow depth measurements remained at 6 lb/ft² (0.29 kN/m²).

The following additional information was also considered when establishing the snow load zones on the map of the United States (Fig. 7.2-1).

1. The number of years of record available at each location;
2. Additional meteorological information available from NWS, National Resources Conservation Service (NRCS) snow surveys (SNOTEL) and other sources. NRCS was formerly known as Soil Conservation Service;

3. Maximum snow loads observed;
4. Regional topography; and
5. The elevation of each location.

The map was updated in the 1995 edition of this standard and was changed in the 2016 edition.

In much of the south, infrequent but severe snowstorms disrupted life in the area to the point that meteorological observations were missed. In these and similar circumstances, more value was given to the statistical values for stations with complete records. Year-by-year checks were made to verify the significance of data gaps.

The mapped snow loads cannot be expected to represent all the local differences that may occur within each zone. Because local differences exist, each zone has been positioned so as to encompass essentially all the statistical values associated with normal sites in that zone. Although the zones represent statistical values, not maximum observed values, the maximum observed values were helpful in establishing the position of each zone.

For sites not covered in Fig. 7.2-1, design values should be established from meteorological information, with consideration given to the orientation, elevation, and records available at each location. The same method can also be used to improve upon the values presented in Fig. 7.2-1. Detailed study of a specific site may generate a design value lower than that indicated by the generalized national map. It is appropriate in such a situation to use the lower value established by the detailed study. Occasionally a detailed study may indicate that a higher design value should be used than the national map indicates. Again, results of the detailed study should be followed.

Using the database used to establish the ground snow loads in Fig. 7.2-1, additional meteorological data, and a methodology that meets the requirements of Section 7.2 (Tobiasson and Greator 1996), ground snow loads have been determined for every town in New Hampshire (Tobiasson et al. 2000, 2002).

The area covered by a site-specific case study varies depending on local climate and topography. In some places, a single case study suffices for an entire community, but in others, varying local conditions limit a "site" to a much smaller area. The area of applicability usually becomes clear as information in the vicinity is examined for the case study.

As suggested by the footnote, it is not appropriate to use only the site-specific information in Table C7.2-1 for design purposes. It lacks an appreciation for surrounding station information and, in a few cases, is based on rather short periods of record. The map or a site-specific case study provides more valuable information.

The importance of conducting detailed studies for locations not covered in Fig. 7.2-1 is shown in Table C7.2-2.

For some locations within the case study (CS) areas of the northeast (Fig. 7.2-1), ground snow loads exceed 100 lb/ft² (4.79 kN/m²). Even in the southern portion of the Appalachian Mountains, not far from sites where a 15-lb/ft² (0.72-kN/m²) ground snow load is appropriate, ground loads exceeding 50 lb/ft² (2.39 kN/m²) may be required. Lake-effect storms create requirements for ground loads in excess of 75 lb/ft² (3.59 kN/m²) along portions of the Great Lakes. In some areas of the Rocky Mountains, ground snow loads exceed 200 lb/ft² (9.58 kN/m²).

Local records and experience should also be considered when establishing design values.

The values in Table 7.2-1 are for specific Alaskan locations only and generally do not represent appropriate design values for

Table C7.2-2 Comparison of Some Site-Specific Values and Zoned Values in Fig. 7.2-1

State	Location	Elevation ft (m)	Zoned Value lb/ft ² (kN/m ²)	Case Study Value ^a lb/ft ² (kN/m ²)
California	Mount Hamilton	4,210 (1,283)	0 to 2,400 ft (732 m) 0 to 3,500 ft (1,067 m)	30 (1.44)
Arizona	Palisade Ranger Station	7,950 (2,423)	5 to 4,600 ft (0.24 to 1,402 m) 10 to 5,000 ft (0.48 to 1,524 m)	120 (5.75)
Tennessee	Monteagle	1,940 (591)	10 to 1,800 ft (0.48 to 549 m)	15 (0.72)
Maine	Sunday River Ski Area	900 (274)	90 to 700 ft (4.31 to 213 m)	100 (4.79)

^aBased on a detailed study of information in the vicinity of each location.

other nearby locations. They are presented to illustrate the extreme variability of snow loads within Alaska. This variability precludes statewide mapping of ground snow loads there. Site-specific case studies were conducted to provide the Alaskan values in Table 7.2-1.

Valuable information on snow loads for the western states is contained in Structural Engineers Association of Northern California (1981), MacKinlay and Willis (1965), Brown (1970), U.S. Department of Agriculture, Soil Conservation Service (1970), Structural Engineers Association of Colorado (2016), Structural Engineers Association of Oregon (2013), Structural Engineers Association of Arizona (1981), Theisen et al. (2004), Structural Engineers Association of Washington (1995), Structural Engineers Association of Utah (1992), Placer County Building Division (1985), Al Hataillah et al. (2015), Curtis and Grimes (2004), and Maji (1999).

Most of these references for the western United States use annual probabilities of being exceeded of 2% (50-year mean recurrence interval). Reasonable, but not exact, factors for converting from other annual probabilities of being exceeded to the value herein are presented in Table C7.2-3. For example, a ground snow load based on a 3.3% annual probability of being exceeded (30-yr mean recurrence interval) should be multiplied by 1.15 to generate a value of p_g for use in Eq. (7.3-1).

The design loads in Table 7.2-2 are based upon a reliability analysis, targeted at a reliability index $\beta = 3.0$, which is consistent with Table 1.3-1. Therefore, unlike the ground snow loads for other states, the mean recurrence interval for the tabulated Colorado snow loads is not necessarily 50 years. However, like all snow loads in the standard, the Colorado snow loads are designed to be used with the load factor of 1.6 for strength-based design per Chapter 2 of this standard. Refer to the cited report from the Structural Engineers Association of Colorado (2016).

Regardless of the methodology used to obtain ground snow loads (e.g., Fig. 7.2-1, a case study or a state study), the ASCE 7 snow load provisions should be used to obtain the ground-to-roof conversion, unbalanced loads, drift loads, and related items.

Table C7.2-3 Factors for Converting from Other Annual Probabilities of Being Exceeded and Other Mean Recurrence Intervals to That Used in This Standard

Annual Probability of Being Exceeded (%)	Mean Recurrence Interval (yr)	Multiplication Factor
10	10	1.82
4	25	1.20
3.3	30	1.15
1	100	0.82

The snow load provisions of several editions of the National Building Code of Canada served as a guide in preparing the snow load provisions in this standard. However, there are some important differences between the Canadian and the United States databases. They include the following:

1. The Canadian ground snow loads are based on a 3.3% annual probability of being exceeded (30-yr mean recurrence interval) generated by using the extreme-value, Type-I (Gumbel) distribution, while the normal-risk values in this standard are based on a 2% annual probability of being exceeded (50-yr mean recurrence interval) generated by a lognormal distribution.
2. The Canadian loads are based on measured depths and regionalized densities based on four or fewer measurements per month. Because of the infrequency of density measurements, an additional weight of rain is added (Newark 1984). In this standard, the weight of the snow is based on many years of frequently measured weights obtained at 204 locations across the United States. Those measurements contain many rain-on-snow events, and thus a separate rain-on-snow surcharge load is not needed except for some roofs with a slope less than $W/50$ as per Section 7.10.

The Importance Factor times the ground snow load is the required balanced snow load for snow accumulation surfaces such as decks, balconies, and subterranean spaces located below the depth of the ground snow. Such snow accumulation surfaces are not subject to the same level of wind erosion or building thermal effect as typical “aboveground” roofs. As such, the flat roof snow load in Eq. (7.3-1) would underestimate the snow load. Note that sliding snow loads and snow drift loads on such surfaces are also possible.

In the 2016 version of the standard, ground snow load contours for selected western states were eliminated from the ground load map in Fig. 7.2-1. They were replaced with tables listing the ground snow load and elevation for select locations in each state. These new western state tables (Tables 7.2-2 through 7.2-8) were developed from detailed state ground snow load studies that satisfy the requirements of this section. Finally, one must also consider any statutory requirements of the Authority Having Jurisdiction. Any such Authority Having Jurisdiction statutory requirements are not included in the state ground snow load tables.

C7.3 FLAT ROOF SNOW LOADS, p_f

The live load reductions in Section 4.8 should not be applied to snow loads. The minimum allowable values of p_f presented in Section 7.3 acknowledge that in some areas a single major storm can generate loads that exceed those developed from an analysis of weather records and snow load case studies.

The factors in this standard that account for the thermal, aerodynamic, and geometric characteristics of the structure in its particular setting were developed using the National Building Code of Canada (National Research Council of Canada 1990) as a point of reference. The case study reports in Peter et al. (1963), Schriever et al. (1967), Lorenzen (1970), Lutes and Schriever (1971), Elliott (1975), Mitchell (1978), Meehan (1979), and Taylor (1979, 1980) were examined in detail.

In addition to these published references, an extensive program of snow load case studies was conducted by eight universities in the United States, the U.S. Army Corps of Engineers’ Alaska District, and the U.S. Army Cold Regions Research and Engineering Laboratory (CRREL) for the Corps of Engineers. The results of this program were used to modify the Canadian methodology to better fit U.S. conditions. Measurements obtained during the severe winters of 1976–1977 and 1977–1978 are included. A statistical analysis of some of that information is presented in O’Rourke et al. (1983). The experience and perspective of many design professionals, including several with expertise in building failure analysis, have also been incorporated.

C7.3.1 Exposure Factor, C_e . Except in areas of “aerodynamic shade,” where loads are often increased by snow drifting, less snow is present on most roofs than on the ground. Loads in unobstructed areas of conventional flat roofs average less than 50% of ground loads in some parts of the country. The values in this standard are above-average values, chosen to reduce the risk of snow load-induced failures to an acceptably low level. Because of the variability of wind action, a conservative approach has been taken when considering load reductions by wind.

The effects of exposure are handled on two scales. First, Eq. (7.3-1) contains a basic exposure factor of 0.7. Second, the type of surface roughness and the exposure of the roof are handled by exposure factor C_e . This two-step procedure generates ground-to-roof load reductions as a function of exposure that range from 0.49 to 0.84.

Table 7.3-1 has been changed from what appeared in a prior version of this standard to separate regional wind issues associated with surface roughness from local wind issues associated with roof exposure. This change was made to better define categories without significantly changing the values of C_e .

Although there is a single “regional” surface roughness category for a specific site, different roofs of a structure may have different exposure factors caused by obstruction provided by higher portions of the structure or by objects on the roof. For example, in surface roughness category C, an upper level roof could be fully exposed ($C_e = 0.9$) while a lower level roof would be partially exposed ($C_e = 1.0$) because of the presence of the upper level roof, as shown in Example 3, this chapter.

The adjective “windswept” is used in the “mountainous areas” surface roughness category to preclude use of this category in those high mountain valleys that receive little wind.

The normal, combined exposure reduction in this standard is 0.70 as compared with a normal value of 0.80 for the ground-to-roof conversion factor in the 1990 National Building Code of Canada. The decrease from 0.80 to 0.70 does not represent decreased safety but arises because of increased choices of exposure and thermal classification of roofs (i.e., five surface roughness categories, three roof exposure categories, and four thermal categories in this standard vs. three exposure categories and no thermal distinctions in the Canadian code).

It is virtually impossible to establish exposure definitions that clearly encompass all possible exposures that exist across the

country. Because individuals may interpret exposure categories somewhat differently, the range in exposure has been divided into several categories rather than just two or three. A difference of opinion of one category results in about a 10% “error” using these several categories and an “error” of 25% or more if only three categories are used.

C7.3.2 Thermal Factor, C_t . Usually, more snow will be present on cold roofs than on warm roofs. An exception to this is discussed in the following text. The thermal condition selected from Table 7.3-2 should represent that which is likely to exist during the life of the structure. Although it is possible that a brief power interruption will cause temporary cooling of a heated structure, the joint probability of this event and a simultaneous peak snow load event is very small. Brief power interruptions and loss of heat are acknowledged in the $C_t = 1.0$ category. Although it is possible that a heated structure will subsequently be used as an unheated structure, the probability of this is rather low. Consequently, heated structures need not be designed for this unlikely event.

Some dwellings are not used during the winter. Although their thermal factor may increase to 1.2 at that time, they are unoccupied, so their Importance Factor reduces to 0.8. The net effect is to require the same design load as for a heated, occupied dwelling.

Discontinuous heating of structures may cause thawing of snow on the roof and subsequent refreezing in lower areas. Drainage systems of such roofs have become clogged with ice, and extra loads associated with layers of ice several inches thick have built up in these undrained lower areas. The possibility of similar occurrences should be investigated for any intermittently heated structure.

Similar icings may build up on cold roofs subjected to meltwater from warmer roofs above. Exhaust fans and other mechanical equipment on roofs may also generate meltwater and icings.

Icicles and ice dams are a common occurrence on cold eaves of sloped roofs. They introduce problems related to leakage and to loads. Large ice dams that can prevent snow from sliding off roofs are generally produced by heat losses from within buildings. Icings associated with solar melting of snow during the day and refreezing along eaves at night are often small and transient. Although icings can occur on cold or warm roofs, roofs that are well insulated and ventilated are not commonly subjected to serious icings at their eaves. Methods of minimizing eave icings are discussed in Grange and Hendricks (1976), Klinge (1978), de Marne (1988), Mackinlay (1988), Tobiasson (1988), and Tobiasson and Buska (1993). Ventilation guidelines to prevent problematic icings at eaves have been developed for attics (Tobiasson et al. 1998) and for cathedral ceilings (Tobiasson et al. 1999).

Because ice dams can prevent load reductions by sliding on some warm ($C_t \leq 1.0$) roofs, the “unobstructed slippery surface” curve in Fig. 7.4-1a now only applies to unventilated roofs with a thermal resistance equal to or greater than $30 \text{ ft}^2 \text{ h } ^\circ\text{F}/\text{Btu}$ ($5.3^\circ\text{C m}^2/\text{W}$) and to ventilated roofs with a thermal resistance equal to or greater than $20 \text{ ft}^2 \text{ h } ^\circ\text{F}/\text{Btu}$ ($3.5^\circ\text{C m}^2/\text{W}$). For roofs that are well insulated and ventilated, see $C_t = 1.1$ in Table 7.3-2.

Glass, plastic, and fabric roofs of continuously heated structures are seldom subjected to much snow load because their high heat losses cause snowmelt and sliding. For such specialty roofs, knowledgeable manufacturers and designers should be consulted. The National Greenhouse Manufacturers Association (1988) recommends use of $C_t = 0.83$ for continuously heated greenhouses and $C_t = 1.00$ for unheated or intermittently heated

greenhouses. They suggest a value of $I_s = 1.0$ for retail greenhouses and $I_s = 0.8$ for all other greenhouses. To qualify as a continuously heated greenhouse, a production or retail greenhouse must have a constantly maintained temperature of 50°F (10°C) or higher during winter months. In addition, it must also have a maintenance attendant on duty at all times or an adequate temperature alarm system to provide warning in the event of a heating system failure. Finally, the greenhouse roof material must have a thermal resistance, R-value, less than $2 \text{ ft}^2 \times h \times ^\circ\text{F}/\text{Btu}$ ($0.4^\circ\text{C m}^2/\text{W}$). In this standard, the C_t factor for such continuously heated greenhouses is set at 0.85. An unheated or intermittently heated greenhouse is any greenhouse that does not meet the requirements of a continuously heated single- or double-glazed greenhouse. Greenhouses should be designed so that the structural supporting members are stronger than the glazing. If this approach is used, any failure caused by heavy snow loads will be localized and in the glazing. This should avert progressive collapse of the structural frame. Higher design values should be used where drifting or sliding snow is expected.

Little snow accumulates on warm air-supported fabric roofs because of their geometry and slippery surface. However, the snow that does accumulate is a significant load for such structures and should be considered.

The combined consideration of exposure and thermal conditions generates ground-to-roof factors that range from a low of 0.49 to a high of 1.01. The equivalent ground-to-roof factors in the 1990 National Building Code of Canada (National Research of Canada) are 0.8 for sheltered roofs, 0.6 for exposed roofs, and 0.4 for exposed roofs in exposed areas north of the tree line, all regardless of their thermal condition.

Sack (1988) and case history experience indicate that the roof snow load on open-air structures (e.g., parking structures and roofs over loading docks) and on buildings intentionally kept below freezing (e.g., freezer buildings) can be larger than the nearby ground snow load. It is thought that this effect is caused by the lack of heat flow up from the “warm” earth for these select groups of structures. Open-air structures are explicitly included with unheated structures. For freezer buildings, the thermal factor is specified to be 1.3 to account for this effect. This value is intended specifically for structures constructed to act as freezer buildings and not those that contain freezer enclosures inside.

C7.3.3 Importance Factor, I_s . The Importance Factor, I_s , has been included to account for the need to relate design loads to the consequences of failure. Roofs of most structures that have normal occupancies and functions are designed with an Importance Factor of 1.0, which corresponds to unmodified use of the statistically determined ground snow load for a 2% annual probability of being exceeded (50-yr mean recurrence interval).

A study of the locations in Table C7.2-1 showed that the ratio of the values for 4% and 2% annual probabilities of being exceeded (the ratio of the 25-yr to 50-yr mean recurrence interval values) averaged 0.80 and had a standard deviation of 0.06. The ratio of the values for 1% and 2% annual probabilities of being exceeded (the ratio of the 100-yr to 50-yr mean recurrence interval values) averaged 1.22 and had a standard deviation of 0.08. On the basis of the nationwide consistency of these values, it was decided that only one snow load map need be prepared for design purposes and that values for lower and higher risk situations could be generated using that map and constant factors.

Lower and higher risk situations are established using the Importance Factors for snow loads in Table 1.5-2. These factors

range from 0.8 to 1.2. The factor 0.8 bases the average design value for that situation on an annual probability of being exceeded of about 4% (about a 25-year mean recurrence interval). The factor 1.2 is nearly that for a 1% annual probability of being exceeded (about a 100-year mean recurrence interval). In the 2016 version of the standard, the Importance Factor is now appropriately applied to the drift height in Fig. 7.6-1. As a result of the functional form, the drift surcharge in pounds per unit cross-wind width ($h_d(4h_d)\gamma/2$) is proportional to the Importance Factor.

C7.3.4 Minimum Snow Load for Low-Slope Roofs, p_m . These minimums account for a number of situations that develop on low-slope roofs. They are particularly important considerations for regions where p_g is 20 lb/ft² (0.96 kN/m²) or less. In such areas, single storm events can result in loading for which the basic ground-to-roof conversion factor of 0.7, as well as the C_e and C_t factors, are not applicable.

The unbalanced load for hip and gable roofs, with an eave to ridge distance W of 20 ft (6.1 m) or less that have simply supported prismatic members spanning from ridge to eave, is greater than or equal to the minimum roof snow load, p_m . Hence, if such a hip and gable roof has a slope that requires unbalanced loading, the minimum snow load would not control and need not be checked for the roof.

C7.4 SLOPED ROOF SNOW LOADS, p_s

Snow loads decrease as the slopes of roofs increase. Generally, less snow accumulates on a sloped roof because of wind action. Also, such roofs may shed some of the snow that accumulates on them by sliding and improved drainage of meltwater. The ability of a sloped roof to shed snow load by sliding is related to the absence of obstructions not only on the roof but also below it, the temperature of the roof, and the slipperiness of its surface. It is difficult to define “slippery” in quantitative terms. For that reason, a list of roof surfaces that qualify as slippery and others that do not is presented in the standard. Most common roof surfaces are on that list. The slipperiness of other surfaces is best determined by comparisons with those surfaces. Some tile roofs contain built-in protrusions or have a rough surface that prevents snow from sliding. However, snow does slide off other smooth-surfaced tile roofs. When a surface may or may not be slippery, the implications of treating it either as a slippery or nonslippery surface should be determined. Because valleys obstruct sliding on slippery surfaced roofs, the dashed lines in Figs. 7.4-1a, b, and c should not be used in such roof areas.

Discontinuous heating of a building may reduce the ability of a sloped roof to shed snow by sliding because meltwater created during heated periods may refreeze on the roof’s surface during periods when the building is not heated, thereby “locking” the snow to the roof.

All these factors are considered in the slope reduction factors presented in Fig. 7.4-1 and are supported by Taylor (1983, 1985), Sack et al. (1987), and Sack (1988). The thermal resistance requirements have been added to the “unobstructed slippery surfaces” curve in Fig. 7.4-1a to prevent its use for roofs on which ice dams often form because ice dams prevent snow from sliding. Mathematically the information in Fig. 7.4-1 can be represented as follows:

1. Warm roofs ($C_t = 1.0$ or less):
 - (a) Unobstructed slippery surfaces with $R \geq 30$ ft²h °F/Btu (5.3°C m²/W) if unventilated and $R \geq 20$ ft²h °F/Btu (3.5°C m²/W) if ventilated:

$$\begin{aligned} 0^\circ\text{--}5^\circ \text{ slope} \quad C_s &= 1.0 \\ 5^\circ\text{--}70^\circ \text{ slope} \quad C_s &= 1.0 - (\text{slope} - 5^\circ)/65 \\ >70^\circ \text{ slope} \quad C_s &= 0 \end{aligned}$$

- (b) All other surfaces:

$$\begin{aligned} 0^\circ\text{--}30^\circ \text{ slope} \quad C_s &= 1.0 \\ 30^\circ\text{--}70^\circ \text{ slope} \quad C_s &= 1.0 - (\text{slope} - 30^\circ)/40^\circ \\ >70^\circ \text{ slope} \quad C_s &= 0 \end{aligned}$$

2. Cold roofs with $C_t = 1.1$

- (a) Unobstructed slippery surfaces:

$$\begin{aligned} 0^\circ\text{--}10^\circ \text{ slope} \quad C_s &= 1.0 \\ 10^\circ\text{--}70^\circ \text{ slope} \quad C_s &= 1.0 - (\text{slope} - 10^\circ)/60 \\ >70^\circ \text{ slope} \quad C_s &= 0 \end{aligned}$$

- (b) All other surfaces:

$$\begin{aligned} 0^\circ\text{--}37.5^\circ \text{ slope} \quad C_s &= 1.0 \\ 37.5^\circ\text{--}70^\circ \text{ slope} \quad C_s &= 1.0 - (\text{slope} - 37.5^\circ)/32.5^\circ \\ >70^\circ \text{ slope} \quad C_s &= 0 \end{aligned}$$

3. Cold roofs ($C_t = 1.2$):

- (a) Unobstructed slippery surfaces:

$$\begin{aligned} 0^\circ\text{--}15^\circ \text{ slope} \quad C_s &= 1.0 \\ 15^\circ\text{--}70^\circ \text{ slope} \quad C_s &= 1.0 - (\text{slope} - 15^\circ)/55^\circ \\ >70^\circ \text{ slope} \quad C_s &= 0 \end{aligned}$$

- (b) All other surfaces:

$$\begin{aligned} 0^\circ\text{--}45^\circ \text{ slope} \quad C_s &= 1.0 \\ 45^\circ\text{--}70^\circ \text{ slope} \quad C_s &= 1.0 - (\text{slope} - 45^\circ)/25^\circ \\ >70^\circ \text{ slope} \quad C_s &= 0 \end{aligned}$$

If the ground (or another roof of less slope) exists near the eave of a sloped roof, snow may not be able to slide completely off the sloped roof. This may result in the elimination of snow loads on upper portions of the roof and their concentration on lower portions. Steep A-frame roofs that nearly reach the ground are subject to such conditions. Lateral and vertical loads induced by such snow should be considered for such roofs.

If the roof has snow retention devices (installed to prevent snow and ice from sliding off the roof), it should be considered an obstructed roof and the slope factor C_s should be based on the “All Other Surfaces” curves in Fig. 7.4-1.

C7.4.3 Roof Slope Factor for Curved Roofs. These provisions were changed from those in the 1993 edition of this standard to cause the load to diminish along the roof as the slope increases.

C7.4.4 Roof Slope Factor for Multiple Folded Plate, Sawtooth, and Barrel Vault Roofs. Because these types of roofs collect extra snow in their valleys by wind drifting and snow creep and sliding, no reduction in snow load should be applied because of slope.

C7.4.5 Ice Dams and Icicles along Eaves. The intent is to consider heavy loads from ice that forms along eaves only for structures where such loads are likely to form. It is also not considered necessary to analyze the entire structure for such loads, just the eaves themselves. Eave ice dam loads with various return periods on roofs with overhangs of 4 ft (1.2 m) or less are presented in O’Rourke et al. (2007).

This provision is intended for short roof overhangs and projections, with a horizontal extent less than 5 ft (1.5 m). In instances where the horizontal extent is greater than 5 ft (1.5 m), the surcharge that accounts for eave ice damming need only extend for a maximum of 5 ft (1.5 m) from the eave of the heated structure (Fig. C7.4-1).

C7.5 PARTIAL LOADING

In many situations, a reduction in snow load on a portion of a roof by wind scour, melting, or snow-removal operations

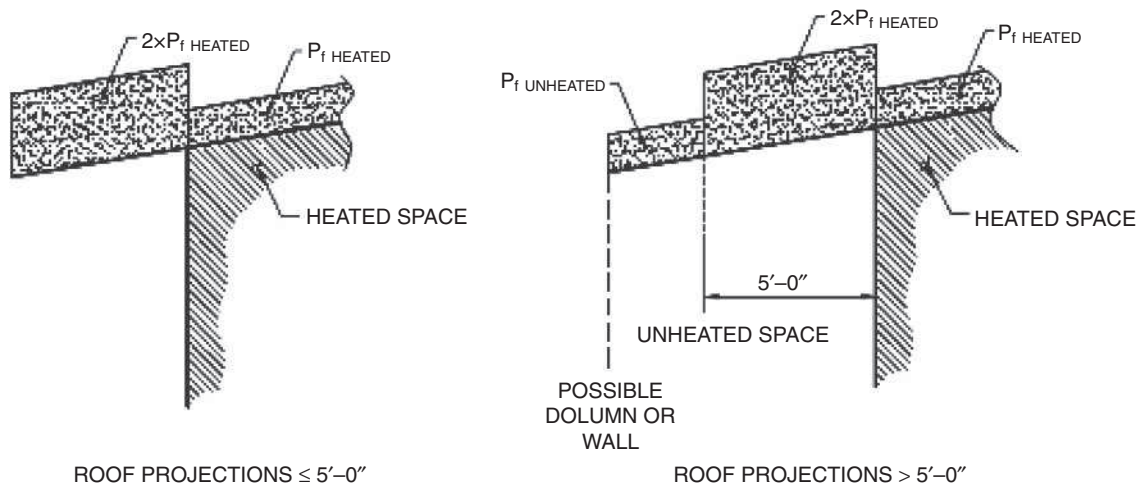


Figure C7.4-1 Eave Ice Dam Loading

simply reduces the stresses in the supporting members. However, in some cases a reduction in snow load from an area induces higher stresses in the roof structure than occur when the entire roof is loaded. Cantilevered roof joists are a good example; removing half the snow load from the cantilevered portion increases the bending stress and deflection of the adjacent continuous span. In continuous beam roof systems, problems can occur during snow-removal operations. The nonuniform loading imposed by the removal of snow load in an indiscriminate manner can significantly alter the load distribution throughout the roof system and can result in marked increases in stresses and deflections over those experienced during uniform loading. In other situations, adverse stress reversals may result.

The simplified provisions offered for continuous beam roof systems have been adopted to mimic real loadings experienced by this common structural system. The Case 1 load scenario simulates a critical condition encountered when, for example, the process of removing a portion of the snow in each span starts at one building end and proceeds toward the other end. The Case 2 load scenario is intended to model loading nonuniformity caused by wind scour or local heat loss and snowmelt at the building edges. The Case 3 load group is intended to encompass conditions that might be encountered, for example, when removal operations must be discontinued and are restarted in a different location, or when multiple snow removal crews start at different roof locations.

The intent is not to require consideration of multiple “checkerboard” loadings.

Members that span perpendicular to the ridge in gable roofs with slopes between 2.38° ($\frac{1}{2}$ on 12) and 30.3° (7 on 12) are exempt from partial load provisions because the unbalanced load provisions of Section 7.6.1 provide for this situation.

C7.6 UNBALANCED ROOF SNOW LOADS

Unbalanced snow loads may develop on sloped roofs because of sunlight and wind. Winds tend to reduce snow loads on windward portions and increase snow loads on leeward portions. Because it is not possible to define wind direction with assurance, winds from all directions should generally be considered when establishing unbalanced roof loads.

C7.6.1 Unbalanced Snow Loads for Hip and Gable Roofs. The expected shape of a gable roof drift is nominally

a triangle located close to the ridgeline. Recent research suggests that the size of this nominally triangular gable roof drift is comparable to a leeward roof step drift with the same fetch. For certain simple structural systems, for example, wood or light-gauge roof rafter systems with either a ridge board or a supporting ridge beam, with small eave to ridge distances, the drift is represented by a uniform load of $I_s \times p_g$ from eave to ridge. For all other gable roofs, the drift is represented by a rectangular distribution located adjacent to the ridge. The location of the centroid for the rectangular distribution is identical to that for the expected triangular distribution. The intensity is the average of that for the expected triangular distribution.

The design snow load on the windward side for the unbalanced case, $0.3p_s$, is based upon case histories presented in Taylor (1979) and O'Rourke and Auren (1997) and discussed in Tobiasson (1999). The lower limit of $\theta = 2.38^\circ$ is intended to exclude low-slope roofs, such as membrane roofs, on which significant unbalanced loads have not been observed. The upper bound of $\theta > 7$ on 12 (30.2°) is intended to exclude high-slope roofs on which significant unbalanced loads have not been observed. That is, although an upper bound for the angle of repose for fresh-fallen snow is about 70° as given in Fig. 7.4-1, the upper bound for the angle of repose of drifted snow is about 30° .

As noted, observed gable roof drifts are nominally triangular in shape. The surcharge is essentially zero at the ridge, and the top surface of the surcharge is nominally horizontal. As such, an upper bound for an actual surcharge atop the sloped roof snow load, p_s , would be a triangular distribution: zero at the ridge and a height at the eave equal to the elevation difference between the eave and the ridge.

For intersecting gable roofs and similar roof geometries, some codes and standards have required a valley drift load. Such valley drift loads are not required in ASCE 7. However, valley locations are subject to unbalanced or gable roof drifts, as described in Section 7.6.1. An example of unbalanced loading on an L-shaped gable roof is presented in O'Rourke (2007).

For intersecting monoslope roofs and intersecting gable roofs with slopes greater than 7 on 12, unbalanced loads are not required in ASCE 7. However, at such valleys, snow on each side of the valley is prevented from sliding by the presence of roof snow on the other side of the valley. As such, the valley portion of the roof (drainage area upslope of the reentrant corner)

is obstructed and the slope factor C_s should be based on the “All Other Surfaces” lines in Fig. 7.4-1.

C7.6.2 Unbalanced Snow Loads for Curved Roofs. The method of determining roof slope is the same as in the 1995 edition of this standard. C_s is based on the actual slope, not an equivalent slope. These provisions do not apply to roofs that are concave upward. For such roofs, see Section C7.13.

C7.6.3 Unbalanced Snow Loads for Multiple Folded Plate, Sawtooth, and Barrel Vault Roofs. A minimum slope of 3/8 in./ft (1.79°) has been established to preclude the need to determine unbalanced loads for most internally drained membrane roofs that slope to internal drains. Case studies indicate that significant unbalanced loads can occur when the slope of multiple gable roofs is as low as 1/2 in./ft (2.38°).

The unbalanced snow load in the valley is $2p_f/C_e$ to create a total unbalanced load that does not exceed a uniformly distributed ground snow load in most situations.

Sawtooth roofs and other “up-and-down” roofs with significant slopes tend to be vulnerable in areas of heavy snowfall for the following reasons:

1. They accumulate heavy snow loads and are therefore expensive to build.
2. Windows and ventilation features on the steeply sloped faces of such roofs may become blocked with drifting snow and be rendered useless.
3. Meltwater infiltration is likely through gaps in the steeply sloped faces if they are built as walls because slush may accumulate in the valley during warm weather. This accumulation can promote progressive deterioration of the structure.
4. Lateral pressure from snow drifted against clerestory windows may break the glass.
5. The requirement that snow above the valley not be at an elevation higher than the snow above the ridge may limit the unbalanced load to less than $2p_f/C_e$.

C7.6.4 Unbalanced Snow Loads for Dome Roofs. This provision is based on a similar provision in the 1990 National Building Code of Canada.

C7.7 DRIFTS ON LOWER ROOFS (AERODYNAMIC SHADE)

When a rash of snow load failures occurs during a particularly severe winter, there is a natural tendency for concerned parties to initiate across-the-board increases in design snow loads. This is generally a technically ineffective and expensive way of attempting to solve such problems because most failures associated with snow loads on roofs are caused not by moderate overloads on every square foot (square meter) of the roof, but rather by localized significant overloads caused by drifted snow.

Drifts accumulate on roofs (even on sloped roofs) in the wind shadow of higher roofs or terrain features. Parapets have the same effect. The affected roof may be influenced by a higher portion of the same structure or by another structure or terrain feature nearby if the separation is 20 ft (6.1 m) or less. When a new structure is built within 20 ft (6.1 m) of an existing structure, drifting possibilities should also be investigated for the existing structure (see Sections C7.7.2 and C7.12). The snow that forms drifts may come from the roof on which the drift forms, from higher or lower roofs, or, on occasion, from the ground.

The leeward drift load provisions are based on studies of snow drifts on roofs (Speck 1984; Taylor 1984; and O’Rourke et al.

1985, 1986). Drift size is related to the amount of driftable snow as quantified by the upwind roof length and the ground snow load. Drift loads are considered for ground snow loads as low as 5 lb/ft² (0.24 kN/m²). Case studies show that, in regions with low ground snow loads, drifts 3 to 4 ft (0.9 to 1.2 m) high can be caused by a single storm accompanied by high winds. The leeward drift height limit of 60% of the lower roof length applies to canopies and other lower level roofs with small horizontal projection from the building wall. It is based upon a 30° angle of repose for drifted snow ($\tan(30^\circ) = \text{drift height/drift length} = 0.577 \sim 0.6$) and is consistent with provisions in Section 7.6.1, which exclude roofs steeper than 7 on 12 from the gable roof drift (unbalanced load) provisions.

A change from a prior (1988) edition of this standard involves the width w when the drift height h_d from Fig. 7.6-1 exceeds the clear height h_c . In this situation, the width of the drift is taken as $4h_d^2/h_c$ with a maximum value of $8h_c$. This drift width relation is based upon equating the cross-sectional area of this drift (i.e., $1/2h_c \times w$) with the cross-sectional area of a triangular drift where the drift height is not limited by h_c (i.e., $1/2h_d \times 4h_d$) as suggested by Zallen (1988). The upper limit of drift width is based on studies by Finney (1939) and Tabler (1975) that suggest that a “full” drift has a rise-to-run of about 1:6.5, and case studies (Zallen 1988) that show observed drifts with a rise-to-run greater than 1:10.

The drift height relationship in Fig. 7.6-1 is based on snow blowing off a high roof upwind of a lower roof. The change in elevation where the drift forms is called a “leeward step.” Drifts can also form at “windward steps.” An example is the drift that forms at the downwind end of a roof that abuts a higher structure there. Fig. 7.7-1 shows “windward step” and “leeward step” drifts.

For situations that have the same amount of available snow (i.e., upper and lower roofs of the same length), the drifts that form in leeward steps are larger than those that form in windward steps. In previous versions of the standard, the windward drift height was given as $1/2h_d$ from Fig. 7.6-1 using the length of the lower roof for l_u . Based upon an analysis of case histories in O’Rourke and De Angelis (2002), a value of 3/4 is now prescribed.

Depending on wind direction, any change in elevation between roofs can be either a windward or leeward step. Thus the height of a drift is determined for each wind direction as shown in Example 3, this chapter, and the larger of the two heights is used as the design drift. The drift height relation in Fig. 7.6-1 was based upon a data set of leeward roof step drifts for which the average upwind fetch distance was about 170 ft (52 m). Unfortunately, the empirical relation provides unrealistic results for very short upwind fetch distances and very low ground snow loads. For example, one would calculate a negative drift height for $l_u = 5$ ft (1.5 m) and $p_g = 5$ psf (0.96 kPa). In prior versions of this load standard, this shortcoming was handled by specifying a minimum l_u of 20 ft (6.1 m). In the 2016 edition, this requirement was modified by noting that h_d need not be taken as greater than $\sqrt{(I_s p_g l_n / 4 \gamma)}$. This new limiting relation was determined by assuming that all the upwind snow is transported by wind and the trapping efficiency (percentage of transported snow that remains in the drift) is 50%.

The drift load provisions cover most, but not all, situations. Finney (1939) and O’Rourke (1989) document a larger drift than would have been expected based on the length of the upper roof. The larger drift was caused when snow on a somewhat lower roof, upwind of the upper roof, formed a drift between those two roofs, allowing snow from the upwind lower roof to be carried up onto the upper roof then into the drift on its downwind side. It was suggested that the sum of the lengths of both roofs could be used to calculate the size of the leeward drift. The situation of two

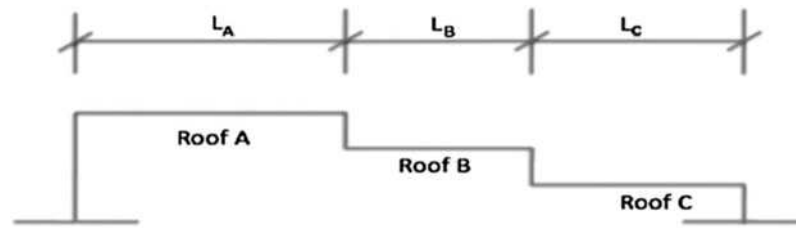


Figure C7.7-1 Roof Steps in Series

roof steps in series was studied by O'Rourke and Kuskowski (2005). For the roof geometry sketched in Fig. C7.7-1, the effective upwind fetch for the leeward drift atop roof C caused by wind from left to right was shown to be $L_B + 0.75L_A$. That is, some of the snow originally on roof A ends up in the leeward drift atop roof B, thereby reducing the amount of snow available for drift formation atop roof C. For the windward snowdrift atop roof B caused by wind from right to left, the effective upwind fetch was shown to be $L_B + 0.85L_C$. The analysis for windward drifts assumed that the elevation difference between roofs B and C was small enough to allow snow initially atop roof C to be blown up onto roof B.

Generally, the addition of a parapet wall on a high roof cannot be relied upon to substantially reduce the leeward snowdrift loading on an adjacent or adjoining lower roof. This is particularly true for the case of a single parapet wall of typical height located at the roof step. Also, the addition of a parapet wall at a roof step would increase the space available for windward drift formation on the lower roof. The issue of potential reduction in leeward drift size at a roof step caused by a parapet wall is discussed in more detail in O'Rourke (2007).

In another situation (Kennedy et al. 1992), a long "spike" drift was created at the end of a long skylight with the wind about 30° off the long axis of the skylight. The skylight acted as a guide or deflector that concentrated drifting snow. This action caused a large drift to accumulate in the lee of the skylight. This drift was replicated in a wind tunnel.

As shown in Fig. 7.7-2, the clear height, h_c , is determined based on the assumption that the upper roof is blown clear of snow in the vicinity of the drift. This assumption is reasonable for windward drifting but does not necessarily hold for leeward drifting. For leeward drifting, the last portion of the upper level roof that would become blown clear of snow is the portion adjacent to the roof step. That is, there may still be snow on the upper level roof when the roof step drift has stopped growing. Nevertheless, for simplicity, the same assumption regarding clear height is used for both leeward and windward drifts.

Tests in wind tunnels (Irwin et al. 1992; Isyumov and Mikitiuk 1992) and flumes (O'Rourke and Weitman 1992) have proven quite valuable in determining patterns of snow drifting and drift loads. For roofs of unusual shape or configuration, wind tunnel or water-flume tests may be needed to help define drift loads. An ASCE standard for wind tunnel testing including procedures to assist in the determination of snow loads on roofs is currently under development.

C7.7.2 Adjacent Structures. One expects a leeward drift to form on an adjacent lower roof only if the lower roof is low enough and close enough to be in the wind shadow (aerodynamic shade) region of the upper roof as sketched in Fig. C7.7-2. The provisions in Section 7.7.2 are based upon a wind shadow region that trails from the upper roof at a 1 downward to 6 horizontal slope.

For windward drifts, the requirements of Section 7.7.1 are to be used. However, the resulting drift may be truncated by eliminating the drift in the horizontal separation region as sketched in Fig. C7.7-3.

C7.7.3 Intersecting Drifts at Low Roofs. The accumulation of drifting snow from perpendicular wind directions can occur concurrently from single or multiple wind and snow events to create an intersecting snowdrift load at reentrant corners. The two resulting drifts are combined as shown in Fig. 7.7-3, where the snowdrift load at the drift intersection is based on the larger drift, not the additive effect of the two drifts (i.e., design h_d at the intersection is h_{d1} or h_{d2} , but not h_{d1} added to h_{d2}). Wind direction at a specific site can change during a single snow event.

C7.8 ROOF PROJECTIONS AND PARAPETS

Drifts around penthouses, roof obstructions, and parapet walls are also of the "windward step" type because the length of the upper roof is small or no upper roof exists. Solar panels, mechanical equipment, parapet walls, and penthouses are examples of roof projections that may cause "windward" drifts on the roof around them. The drift load provisions in Sections 7.7 and 7.8 cover most of these situations adequately, but flat-plate solar collectors may warrant some additional attention. Roofs equipped with several rows of them are subjected to additional snow loads. Before the collectors were installed, these roofs may have sustained minimal snow loads, especially if they were windswept. First, because a roof with collectors is apt to be somewhat "sheltered" by the collectors, it seems appropriate to assume that the roof is partially exposed and calculate a uniform snow load for the entire area as though the collectors did not exist. Second, the extra snow that might fall on the collectors and then slide onto the roof should be computed using the "All Other Surfaces" curve in Fig. 7.4-1b. This value should be applied as a uniform load on the roof at the base of each collector over an area about 2 ft (0.6 m) wide along the length of the collector. The uniform load combined with the load at the base of each collector probably represents a reasonable design load for such situations, except in very windy areas, where extensive snow drifting is to be expected among the collectors. By elevating collectors at least 2 ft (0.61 m) above the height of the balanced snow on an open system of structural supports, the potential for drifting will be diminished significantly. Finally, the collectors should be designed to sustain a load calculated by using the "Unobstructed Slippery Surfaces" curve in Fig. 7.4-1a. This last load should not be used in the design of the roof because the heavier load of sliding snow from the collectors has already been considered. The influence of solar collectors on snow accumulation is discussed in Corotis et al. (1979) and O'Rourke (1979).

Refer to Section C7.7 for more description of the effects that a parapet wall at a high roof can have on the snowdrift loading at an adjacent or adjoining lower roof.

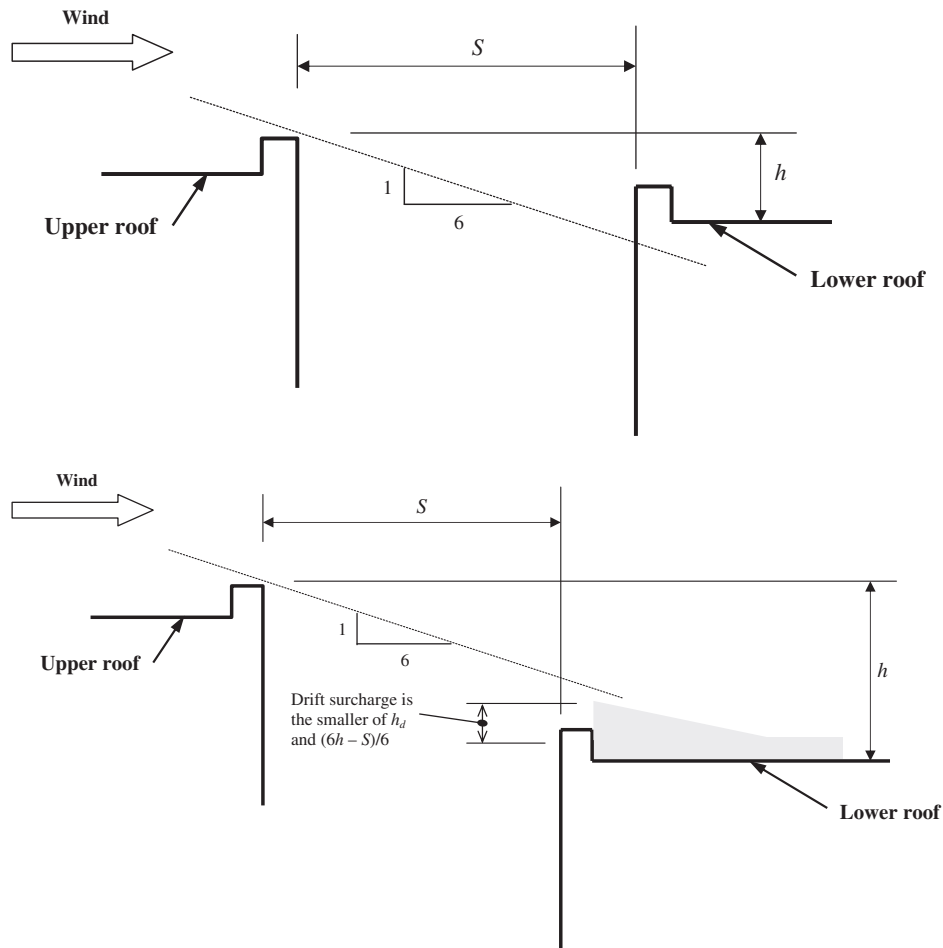


FIGURE C7.7-2 Leeward Snow Drift on Adjacent Roof, Separation $S < 20$ ft (top) Elevation View, $S \geq 6H$; Lower Roof above Wind Shadow (Aerodynamic Shade) Region, No Leeward Drift on Lower Roof. (bottom) Elevation View, $S < 6H$; Lower Roof within Wind Shadow (Aerodynamic Shade) Region, Leeward Drift on Lower Roof; Drift Length Is the Smaller of $(6H - S)$ and $6H_D$

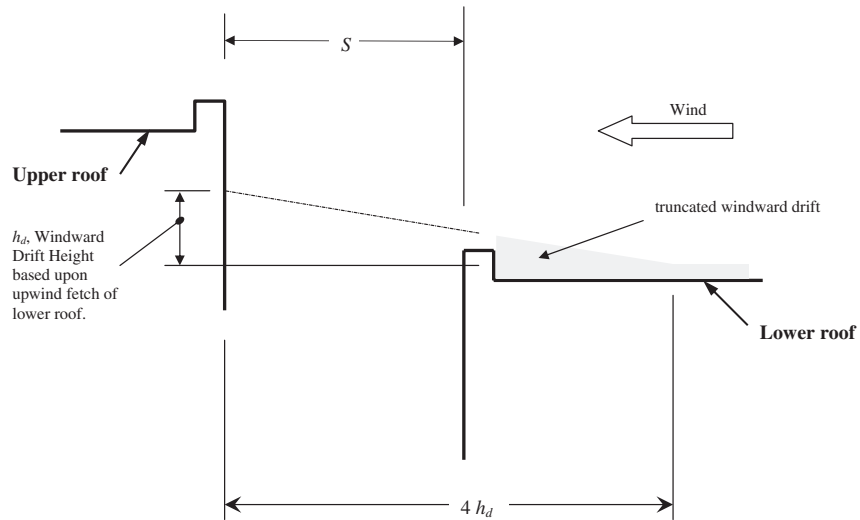


FIGURE C7.7-3 Windward Snow Drift on Adjacent Roof, Separation $S < 20$ ft

The accumulation of drifting snow from perpendicular wind directions can occur concurrently from single or multiple wind and snow events to create an intersecting snowdrift load at parapet wall corners and intersecting roof projections. Where

the two drifts intersect, the drift loads need not be superimposed to create a combined (additive) drift height at the drift intersection. The individual drift loads are based on windward or leeward drifting, as indicated in Section 7.8, and the only difference with

intersecting drifts is the determination of the governing drift load where the two drifts come together.

C7.9 SLIDING SNOW

Situations that permit snow to slide onto lower roofs should be avoided (Paine 1988). Where this is not possible, the extra load of the sliding snow should be considered. Roofs with little slope have been observed to shed snow loads by sliding.

The final resting place of any snow that slides off a higher roof onto a lower roof depends on the size, position, and orientation of each roof (Taylor 1983). Distribution of sliding loads might vary from a uniform 5-ft (1.5-m) wide load, if a significant vertical offset exists between the two roofs, to a 20-ft (6.1-m) wide uniform load, where a low-slope upper roof slides its load onto a second roof that is only a few feet (about a meter) lower or where snowdrifts on the lower roof create a sloped surface that promotes lateral movement of the sliding snow.

In some instances, a portion of the sliding snow may be expected to slide clear of the lower roof. Nevertheless, it is prudent to design the lower roof for a substantial portion of the sliding load to account for any dynamic effects that might be associated with sliding snow.

Snow retention devices are needed on some roofs to prevent roof damage and eliminate hazards associated with sliding snow (Tobiasson et al. 1996). When snow retention devices are added to a sloping roof, snow loads on the roof can be expected to increase. Thus, it may be necessary to strengthen a roof before adding snow retention devices. When designing a roof that will likely need snow retention devices in the future, it may be appropriate to use the “All Other Surfaces” curves in Fig. 7.4-1, not the “Unobstructed Slippery Surfaces” curves. The design of snow retention devices, their anchorages, and their supporting elements should consider the total maximum design roof snow load upslope of the snow retention device (including sloped roof snow load, drift surcharge, unbalanced loads, and sliding snow loads) and the resultant force caused by the roof slope. See Tobiasson et al. (1996) for further information on the effects of roof slope on the snow retention device force. Snow retention devices, their anchorages, and their supporting structural elements should consider the tributary snow loads from a trapezoidal-shaped area with a boundary at the edge of the snow retention device extending upslope at 45 degrees from the vertical to the ridge (Fig. C7.9-1) or to where the tributary area meets an adjacent tributary area.

Friction between the snow and the roof surface is typically taken as zero when calculating the resultant load on the snow retention device.

Sliding snow should also be considered at changes in roof slope where a higher and steeper roof meets a lower, less steep roof (where the difference in slope is greater than 2 on 12). Where this condition occurs, the low roof prevents the high roof from completely unloading because the low roof slides more gradually or not at all. In this case, the accumulated snow at the change in roof slope exceeds p_f of either roof.

When designing the structural elements of the building at the change in roof slope, a rational approach would be to take the sliding snow ($0.4p_fW$) and distribute it a distance of $W/4$ each side of the change in roof slope, accumulating it with the distributed design snow p_f .

C7.10 RAIN-ON-SNOW SURCHARGE LOAD

The ground snow load measurements on which this standard is based contain the load effects of light rain on snow. However, because heavy rains percolate down through snow packs and may drain away, they might not be included in measured values. Where p_g is greater than 20 lb/ft² (0.96 kN/m²), it is assumed that the full rain-on-snow effect has been measured and a separate rain-on-snow surcharge is not needed. The temporary roof load contributed by a heavy rain may be significant. Its magnitude depends on the duration and intensity of the design rainstorm, the drainage characteristics of the snow on the roof, the geometry of the roof, and the type of drainage provided. Loads associated with rain on snow are discussed in Colbeck (1977a, b) and O’Rourke and Downey (2001).

O’Rourke and Downey (2001) show that the surcharge from rain-on-snow loading is an increasing function of eave to ridge distance and a decreasing function of roof slope. That is, rain-on-snow surcharges are largest for wide, low-sloped roofs. The minimum slope reflects that functional relationship.

The following example illustrates the evaluation of the rain-on-snow surcharge. Consider a monoslope roof with slope of 1/4 on 12 and a width of 100 ft (30.5 m) with $C_e = 1.0$, $C_t = 1.1$, $I = 1.2$, and $p_g = 15$ psf (0.72 kN/m²). Because $C_s = 1.0$ for a slope of 1/4 on 12, $p_s = 0.7(1.0)(1.1)(1.0)(1.2)(15) = 14$ psf (0.67 kN/m²). Because the roof slope 1.19° is less than $100/50 = 2.0$, the 5 psf (0.24 kN/m²) surcharge is added to p_s , resulting in a design load of 19 psf (0.91 kN/m²). Because the slope is less than 15°, the minimum load from 7.34 is

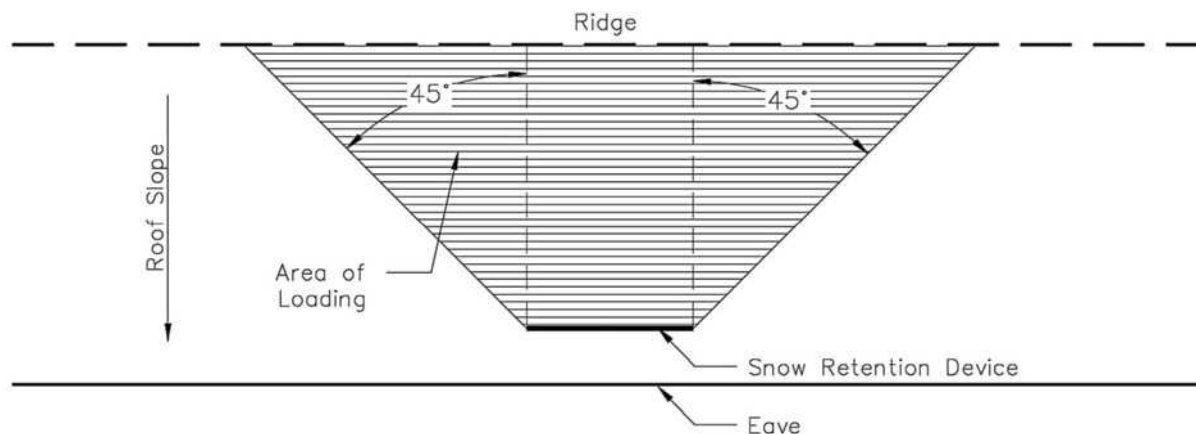


FIGURE C7.9-1 Plan View of Tributary Snow Loads for Snow Retention Device

$I \times p_g = 1.2(15) = 18 \text{ psf}$ (0.86 kN/m^2). Hence, the rain on snow modified load controls.

C7.11 PONDING INSTABILITY

Where adequate slope to drain does not exist, or where drains are blocked by ice, snow meltwater and rain may pond in low areas. Intermittently heated structures in very cold regions are particularly susceptible to blockages of drains by ice. A roof designed without slope or one sloped with only $1/8 \text{ in./ft}$ (0.6°) to internal drains probably contains low spots away from drains by the time it is constructed. When a heavy snow load is added to such a roof, it is even more likely that undrained low spots exist. As rainwater or snow meltwater flows to such low areas, these areas tend to deflect increasingly, allowing a deeper pond to form. If the structure does not possess enough stiffness to resist this progression, failure by localized overloading can result. This mechanism has been responsible for several roof failures under combined rain and snow loads.

It is very important to consider roof deflections caused by snow loads when determining the likelihood of ponding instability from rain-on-snow or snow meltwater.

Internally drained roofs should have a slope of at least $1/4 \text{ in./ft}$ (1.19°) to provide positive drainage and to minimize the chance of ponding. Slopes of $1/4 \text{ in./ft}$ (1.19°) or more are also effective in reducing peak loads generated by heavy spring rain on snow. Further incentive to build positive drainage into roofs is provided by significant improvements in the performance of waterproofing membranes when they are sloped to drain.

Rain loads and ponding instability are discussed in detail in Chapter 8.

C7.12 EXISTING ROOFS

Numerous existing roofs have failed when additions or new buildings nearby caused snow loads to increase on the existing roof. A prior (1988) edition of this standard mentioned this issue only in its commentary where it was not a mandatory provision. The 1995 edition moved this issue to the standard.

The addition of a gable roof alongside an existing gable roof as shown in Fig. C7.12-1 most likely explains why some such metal buildings failed in the south during the winter of 1992–1993. The change from a simple gable roof to a multiple folded plate roof increased loads on the original roof, as would be expected from Section 7.6.3. Unfortunately, the original roofs were not strengthened to account for these extra loads, and they collapsed.

If the eaves of the new roof in Fig. C7.12-1 had been somewhat higher than the eaves of the existing roof, the exposure factor C_e for the original roof may have increased, thereby increasing snow loads on it. In addition, drift loads and loads from sliding snow would also have to be considered.

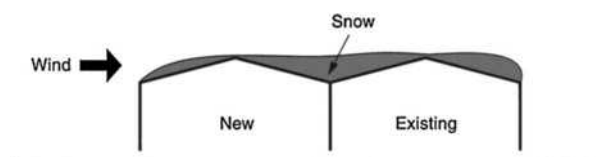


FIGURE C7.12-1 Valley in Which Snow Will Drift Is Created When New Gable Roof Is Added alongside Existing Gable Roof

C7.13 SNOW ON OPEN-FRAME EQUIPMENT STRUCTURES

Snow loads should be considered on all levels of the open-frame structures that can retain snow. The snow accumulations on the flooring, equipment, cable trays, and pipes should be considered. Snow corning effects impacting the accumulation of snow on surfaces such as grating, pipes, cable trays, and equipment should also be considered. Grating is considered to retain snow because of the corning effect of the snow between the grates.

C7.13.2 Snow at Levels below the Top Level. In the absence of site-specific information, the length of the loaded zone is approximated for convenience as the vertical distance to the covering level above.

The drift snow at levels below the top levels caused by wind walls may be ignored because of the limited snow accumulation on such levels.

C7.13.3 Snow Loads on Pipes and Cable Trays. The snow loading on any pipe rack or pipe bridge occurs because of the snow loading on the pipes and cable trays at each level. For pipe racks and pipe bridges wider than 12 ft (4 m) where the spaces between individual pipes are less than the pipe diameter (including insulation), Section 7.13.1 and 7.13.2 should be applied.

C7.13.4 Snow Loads on Equipment and Equipment Platforms. Extended out of service conditions for equipment shall be considered for the loading criteria, particularly equipment that is often out of service for the whole winter.

C7.14 OTHER ROOFS AND SITES

Wind tunnel model studies, similar tests using fluids other than air, for example, water flumes, and other special experimental and computational methods have been used with success to establish design snow loads for other roof geometries and complicated sites (Irwin et al. 1992; Isyumov and Mikitiuk 1992; O'Rourke and Weitman 1992). To be reliable, such methods must reproduce the mean and turbulent characteristics of the wind and the manner in which snow particles are deposited on roofs and then redistributed by wind action. Reliability should be demonstrated through comparisons with situations for which full-scale experience is available.

Examples. The following three examples illustrate the method used to establish design snow loads for some of the situations discussed in this standard. Additional examples are found in O'Rourke and Wrenn (2004).

Example 1: Determine balanced and unbalanced design snow loads for an apartment complex in a suburb of Hartford, Connecticut. Each unit has an 8-on-12 slope unventilated gable roof. The building length is 100 ft (30.5 m), and the eave to ridge distance, W , is 30 ft (9.1 m). Composition shingles clad the roofs. Trees will be planted among the buildings.

Flat Roof Snow Load:

$$p_f = 0.7C_eC_tI_s p_g$$

where

$p_g = 30 \text{ lb/ft}^2$ (1.44 kN/m^2) (from Fig. 7.2-1),

$C_e = 1.0$ (from Table 7.3-1 for Surface Roughness Category B and a partially exposed roof),

$C_t = 1.0$ (from Table 7.3-2), and

$I_s = 1.0$ (from Table 1.5-2).

Thus, $p_f = (0.7)(1.0)(1.0)(1.0)(30) = 21 \text{ lb/ft}^2$ (balanced load), (in SI, $p_f = (0.7)(1.0)(1.0)(1.0)(1.44) = 1.01 \text{ kN/m}^2$).

Because the roof slope is greater than 15° , the minimum roof snow load, p_m , does not apply (see Section 7.3.4).

Sloped-Roof Snow Load:

$$p_s = C_s p_f \quad \text{where } C_s = 1.0$$

(using the “All Other Surfaces” [or solid] line, Fig. 7.4-1a).
Thus, $p_s = 1.00(21) = 19 \text{ lb/ft}^2$
(in SI, $p_s = 1.00(1.01) = 0.92 \text{ kN/m}^2$).

Unbalanced Snow Load: Because the roof slope is greater than $1/2$ on 12 (2.38°), unbalanced loads must be considered. For $p_g = 30 \text{ psf}$ (1.44 kN/m^2) and $W = l_u = 30 \text{ ft}$ (9.14 m), $h_d = 1.86 \text{ ft}$ (0.56 m) from Fig. 7.6-1 and $\gamma = 17.9 \text{ pcf}$ (2.80 kN/m^3) from Eq. (7.7-1). For a 6 on 12 roof, $S = 2.0$ and hence the intensity of the drift surcharge, $h_d \gamma / \sqrt{S}$, is 23.5 psf (1.31 kN/m^2), and its horizontal extent $8\sqrt{S}h_d/3$ is 7.0 ft (2.14 m).

Rain-on-Snow Surcharge: A rain-on-snow surcharge load need not be considered because $p_g > 20 \text{ psf}$ (0.96 kN/m^2) (see Section 7.10). See Fig. C7.14-1 for both loading conditions.

Example 2: Determine the roof snow load for a vaulted theater that can seat 450 people, planned for a suburb of Chicago, Illinois. The building is the tallest structure in a recreation-shopping complex surrounded by a parking lot. Two large deciduous trees are located in an area near the entrance. The building has an 80-ft (24.4-m) span and 15-ft (4.6-m) rise circular arc structural concrete roof covered with insulation and aggregate surfaced built-up roofing. The unventilated roofing system has a thermal resistance of $20 \text{ ft}^2 \text{ hr}^\circ\text{F/Btu}$ ($3.5 \text{ Km}^2/\text{W}$). It is expected that the structure will be exposed to winds during its useful life.

Flat Roof Snow Load:

$$p_f = 0.7 C_e C_t I_s p_g$$

where

$p_g = 25 \text{ lb/ft}^2$ (1.20 kN/m^2) (from Fig. 7.2-1),
 $C_e = 0.9$ (from Table 7.3-1 for Surface Roughness Category B and a fully exposed roof),
 $C_t = 1.0$ (from Table 7.3-2), and
 $I_s = 1.1$ (from Table 1.5-2).

Thus, $p_f = (0.7)(0.9)(1.0)(1.1)(25) = 17 \text{ lb/ft}^2$. In SI, $p_f = (0.7)(0.9)(1.0)(1.1)(1.19) = 0.83 \text{ kN/m}^2$.

Tangent of vertical angle from eaves to crown = $15/40 = 0.375$.

Angle = 21° .

Because the vertical angle exceeds 10° , the minimum roof snow load, p_m , does not apply. See Section 7.3.4.

Sloped-Roof Snow Load:

$$p_s = C_s p_f$$

From Fig. 7.4-1a, $C_s = 1.0$ until slope exceeds 30° , which (by geometry) is 30 ft (9.1 m) from the centerline. In this area, $p_s = 17(1) = 17 \text{ lb/ft}^2$ (in SI $p_s = 0.83(1) = 0.83 \text{ kN/m}^2$). At the eaves, where the slope is (by geometry) 41° , $C_s = 0.72$ and $p_s = 17(0.72) = 12 \text{ lb/ft}^2$ (in SI $p_s = 0.83(0.72) = 0.60 \text{ kN/m}^2$). Because slope at the eaves is 41° , Case II loading applies.

Unbalanced Snow Load: Because the vertical angle from the eaves to the crown is greater than 10° and less than 60° , unbalanced snow loads must be considered.

Unbalanced load at crown = $0.5 p_f = 0.5(17) = 9 \text{ lb/ft}^2$
(in SI, $= 0.5(0.83) = 0.41 \text{ kN/m}^2$).

Unbalanced load at 30° point = $2 p_f C_s / C_e = 2(17)(1.0) / 0.9 = 38 \text{ lb/ft}^2$

(in SI, $= 2(0.83)(1.0) / 0.9 = 1.84 \text{ kN/m}^2$).

Unbalanced load at eaves = $2(17)(0.72) / 0.9 = 27 \text{ lb/ft}^2$
(in SI, $= 2(0.83)(0.72) / 0.9 = 1.33 \text{ kN/m}^2$).

Rain-on-Snow Surcharge: A rain-on-snow surcharge load need not be considered because $p_g > 20 \text{ psf}$ (0.96 kN/m^2) (see Section 7.10). See Fig. C7.14-2 for both loading conditions.

Example 3: Determine design snow loads for the upper and lower flat roofs of a building located where $p_g = 40 \text{ lb/ft}^2$ (1.92 kN/m^2). The elevation difference between the roofs is 10 ft (3 m). The $100 \text{ ft} \times 100 \text{ ft}$ ($30.5 \text{ m} \times 30.5 \text{ m}$) unventilated high portion is heated, and the 170 ft (51.8 m) wide, 100 ft (30.5 m) long low portion is an unheated storage area. The building is in an industrial park in flat open country with no trees or other structures offering shelter.

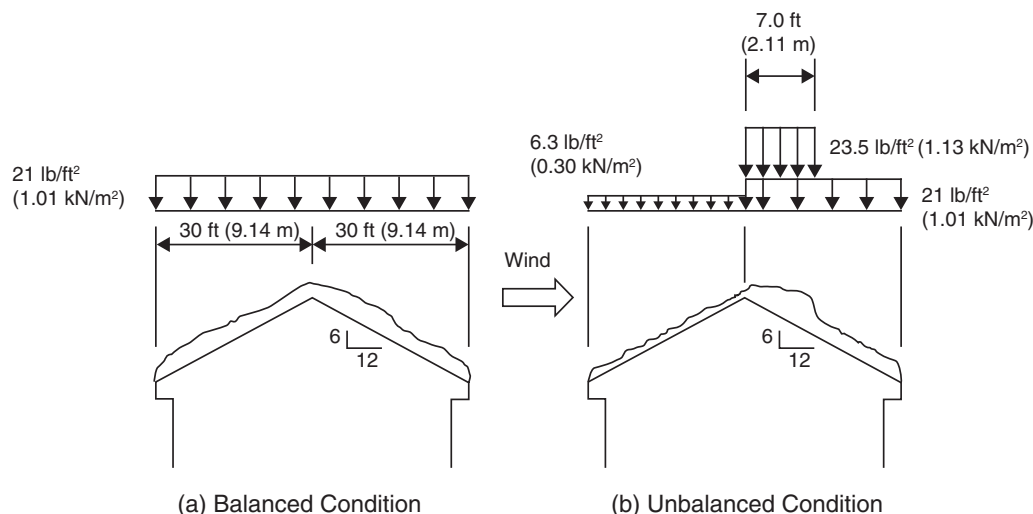


FIGURE C7.14-1 Design Snow Loads for Example 1

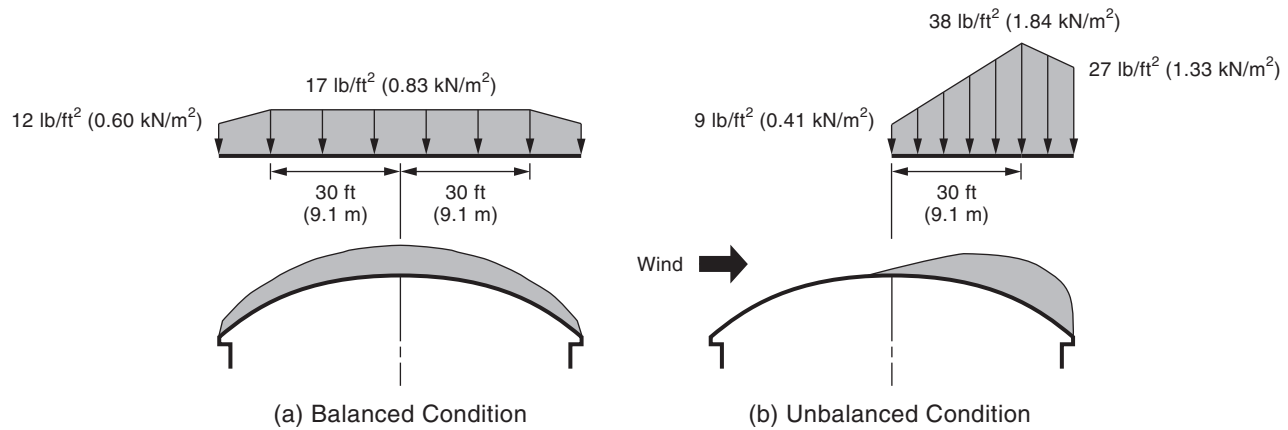


FIGURE C7.14-2 Design Snow Loads for Example 2

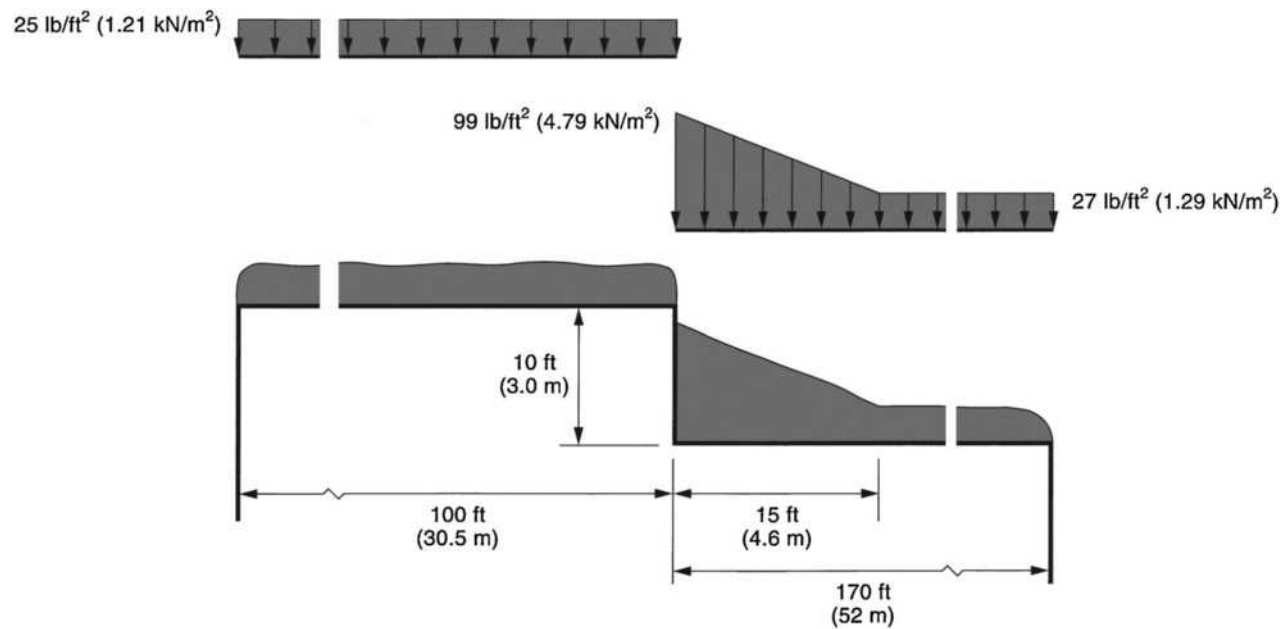


FIGURE C7.14-3 Design Snow Loads for Example 3

High Roof:

$$p_f = 0.7C_eC_tI_s p_g$$

where

$p_g = 40 \text{ lb/ft}^2$ (1.92 kN/m²) (given),
 $C_e = 0.9$ (from Table 7.3-1),
 $C_t = 1.0$ (from Table 7.3-2), and
 $I_s = 1.0$ (from Table 1.5-2).

Thus, $p_f = 0.7(0.9)(1.0)(1.0)(40) = 25 \text{ lb/ft}^2$
 In SI, $p_f = 0.7(0.9)(1.0)(1.0)(1.92) = 1.21 \text{ kN/m}^2$.

Because $p_g = 40 \text{ lb/ft}^2$ (1.92 kN/m²) and $I_s = 1.0$, the minimum roof snow load value of $p_m = 20(1.0) = 20 \text{ lb/ft}^2$ (0.96 kN/m²) and hence does not control (see Section 7.3.4).

Low Roof:

$$p_f = 0.7C_eC_tI_s p_g$$

where

$p_g = 40 \text{ lb/ft}^2$ (1.92 kN/m²) (given),
 $C_e = 1.0$ (from Table 7.3-1) partially exposed because of the presence of a high roof,
 $C_t = 1.2$ (from Table 7.3-2), and
 $I_s = 0.8$ (from Table 1.5-2).

Thus, $p_f = 0.7(1.0)(1.2)(0.8)(40) = 27 \text{ lb/ft}^2$
 In SI, $p_f = 0.7(1.0)(1.2)(0.8)(1.92) = 1.29 \text{ kN/m}^2$.

Because $p_g = 40 \text{ lb/ft}^2$ (1.92 kN/m²) and $I_s = 0.8$, the minimum roof snow load value of $p_m = 20(0.8) = 16 \text{ lb/ft}^2$ (0.77 kN/m²) and hence does not control (see Section 7.3.4).

Drift Load Calculation:

$\gamma = 0.13(40) + 14 = 19 \text{ lb/ft}^3$
 In SI, $\gamma = 0.426(1.92) + 2.2 = 3.02 \text{ kN/m}^3$.

$$h_b = p_f / 19 = 27 / 19 = 1.4 \text{ ft}$$

In SI, $h_b = 1.29 / 3.02 = 0.43 \text{ m}$.

$$h_c = 10 - 1.4 = 8.6 \text{ ft}$$

In SI, $h_c = 3.05 - 0.43 = 2.62 \text{ m}$.

$$h_c / h_b = 8.6 / 1.4 = 6.1$$

In SI, $h_c / h_b = 2.62 / 0.43 = 6.1$.

Because $h_c / h_b \geq 0.2$, drift loads must be considered (see Section 7.7.1).

$$h_d \text{ (leeward step)} = 3.8 \text{ ft (1.16 m)}$$

(Fig. 7.6-1 with $p_g = 40 \text{ lb/ft}^2$ [1.92 kN/m^2]
and $l_u = 100 \text{ ft}$ [30.5 m])

$$h_d \text{ (windward step)} = 3/4 \times 4.8 \text{ ft (1.5 m)}$$

$$= 3.6 \text{ ft (1.1 m) (4.8 ft [1.5 m])}$$

from Fig. 7.6-1 with $p = 40 \text{ lb/ft}^2$ [1.92 kN/m^2]
and l_u = length of lower roof = 170 ft [52 m])

Leeward drift governs, use $h_d = 3.8 \text{ ft (1.16 m)}$
Because $h_d < h_c$,

$$h_d = 3.8 \text{ ft (1.16 m)}$$

$$w = 4h_d = 15.2 \text{ ft (4.64 m), say, } 15 \text{ ft (4.6 m)}$$

$$p_d = h_d \gamma = 3.8(19) = 72 \text{ lb/ft}^2$$

In SI, $p_d = 1.16(3.02) = 3.50 \text{ kN/m}^2$

Rain-on-Snow Surcharge: A rain-on-snow surcharge load need not be considered because p_g is greater than 20 lb/ft^2 (0.96 kN/m^2). See Fig. C7.14-3 for snow loads on both roofs.

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